

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 940+500

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – MNB

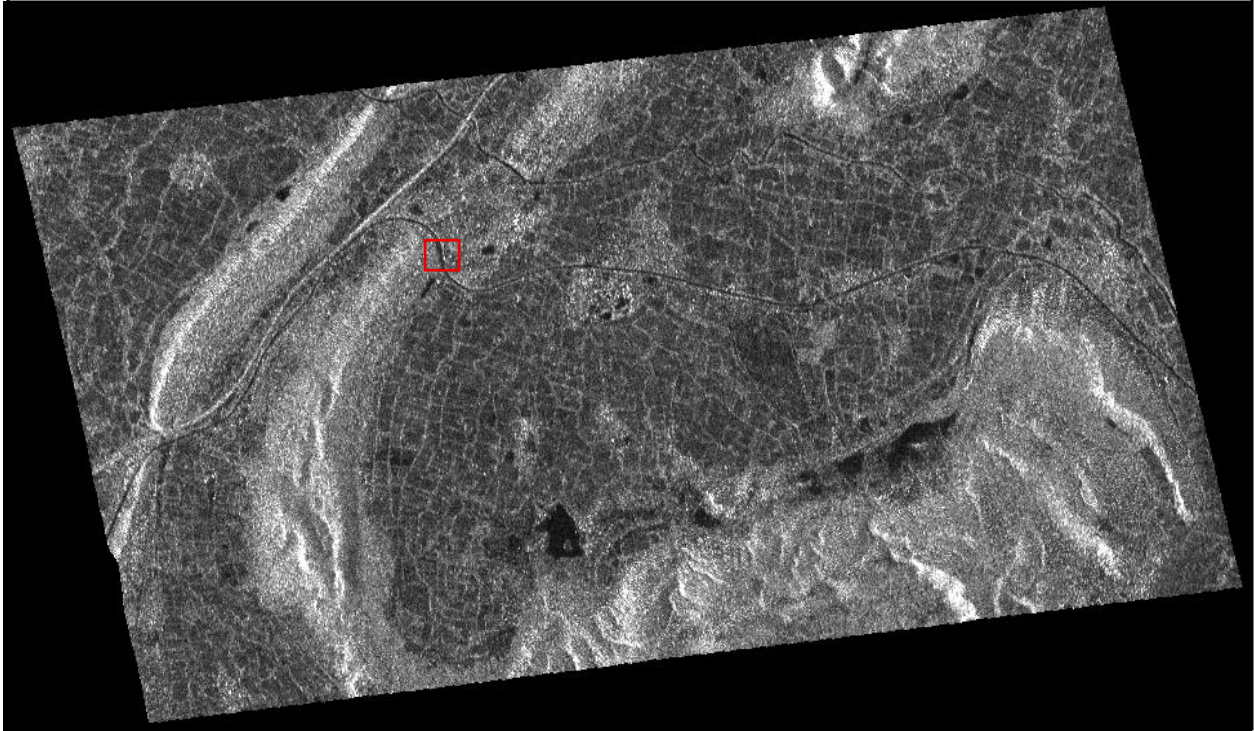
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

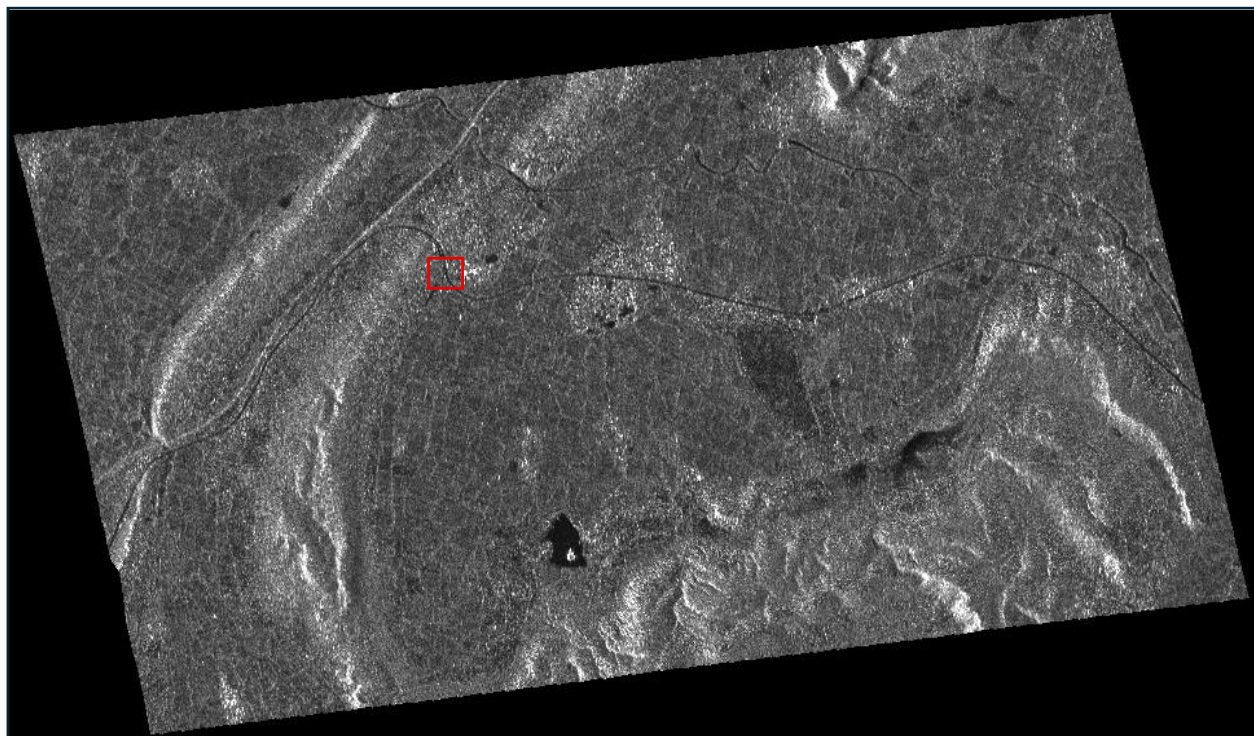
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

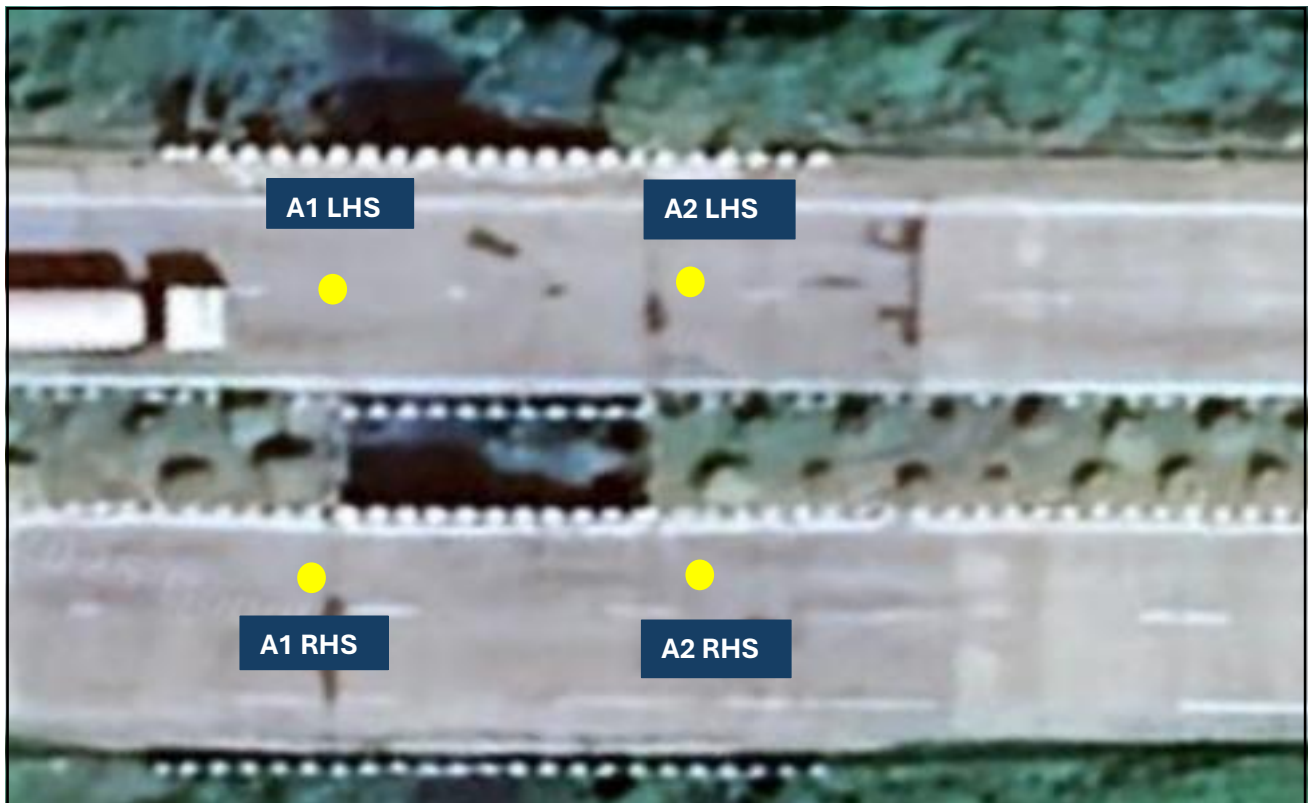


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

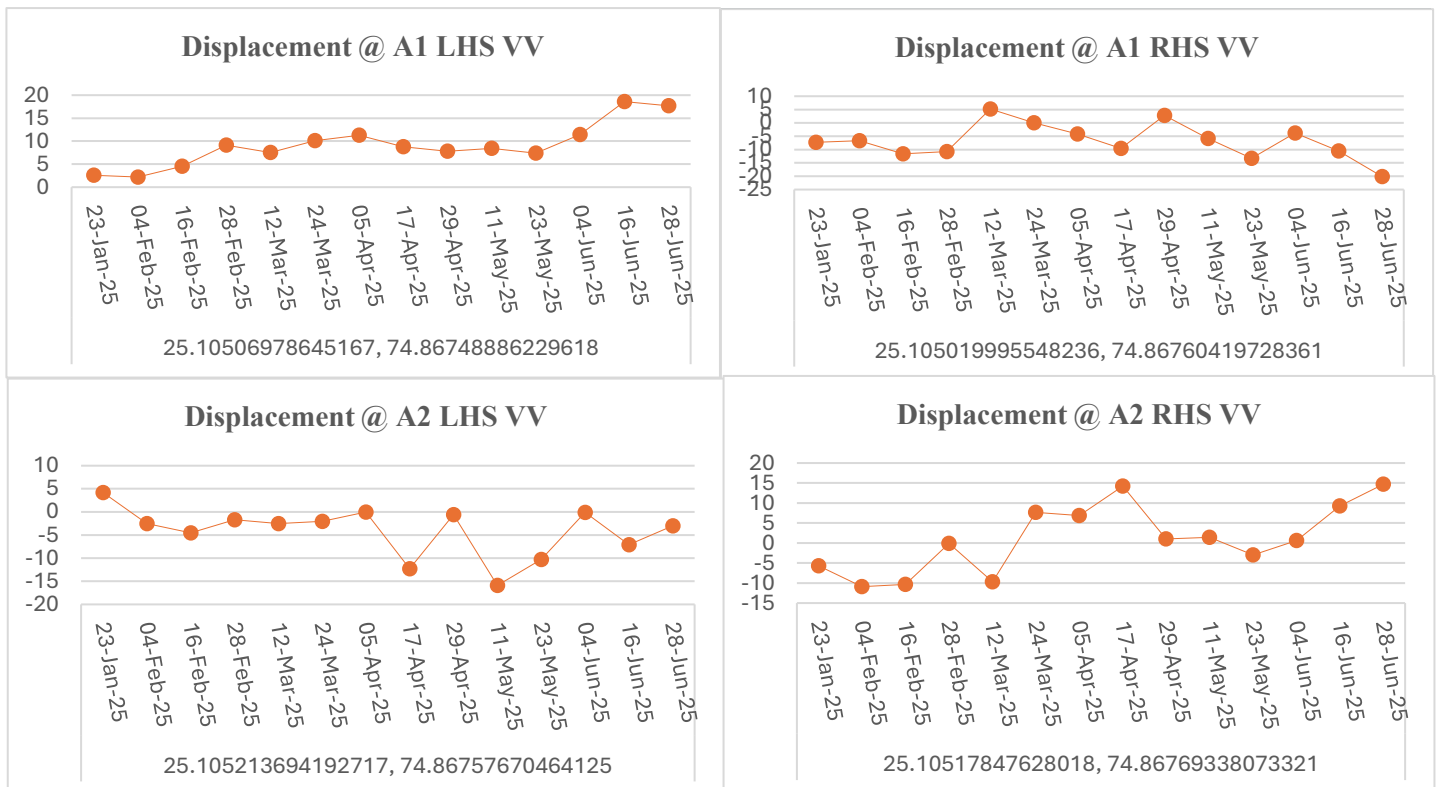
MNB – CH-940+500

Length – 32.74M

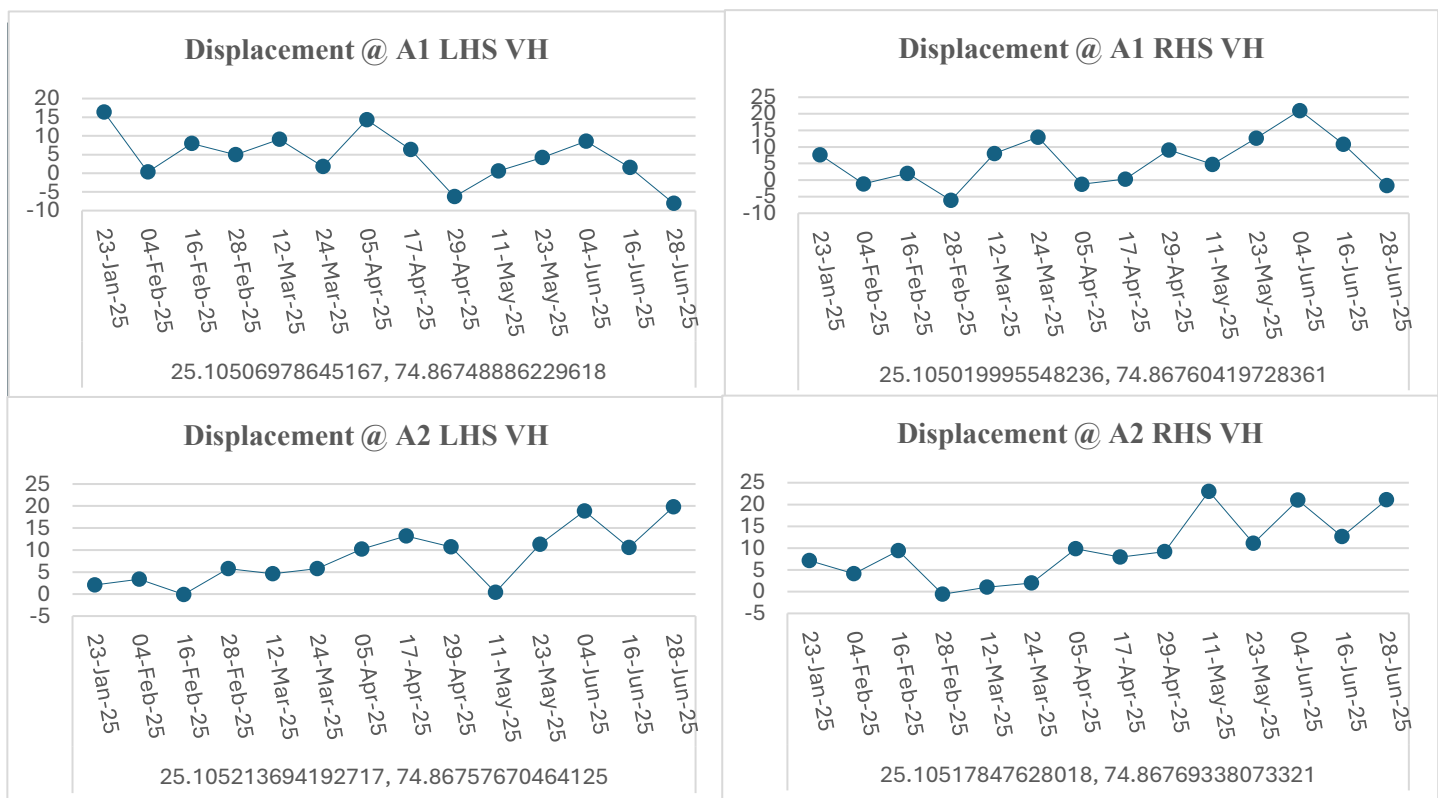


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



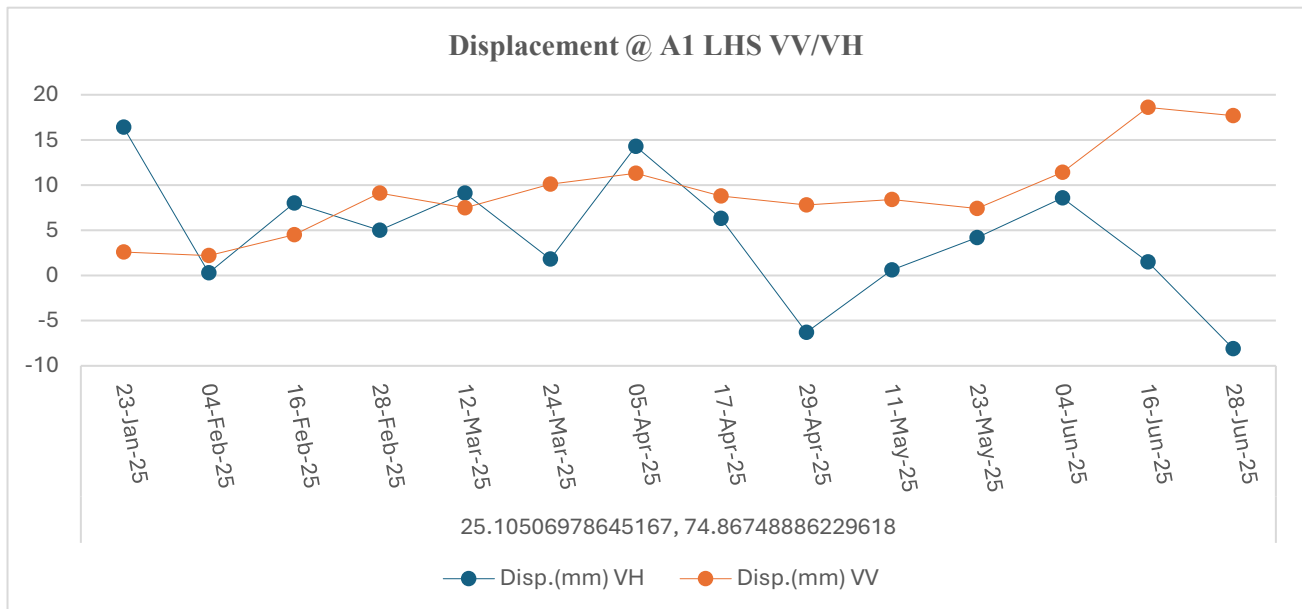
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.10506978645167, 74.86748886229618)



Key Observations:

Displacement Magnitude Patterns:

- Vertical Displacement (VV)** remains consistently positive, generally ranging from +3 mm to +18 mm, indicating a steady upward movement over the entire monitoring period. The magnitude gradually increases, reflecting progressive vertical displacement with no downward settlement phase.
- Horizontal Displacement (VH)** shows significant variability, fluctuating between -7 mm and +17 mm. The behaviour includes alternating positive and negative shifts, indicating cyclic lateral movement rather than a unidirectional trend.

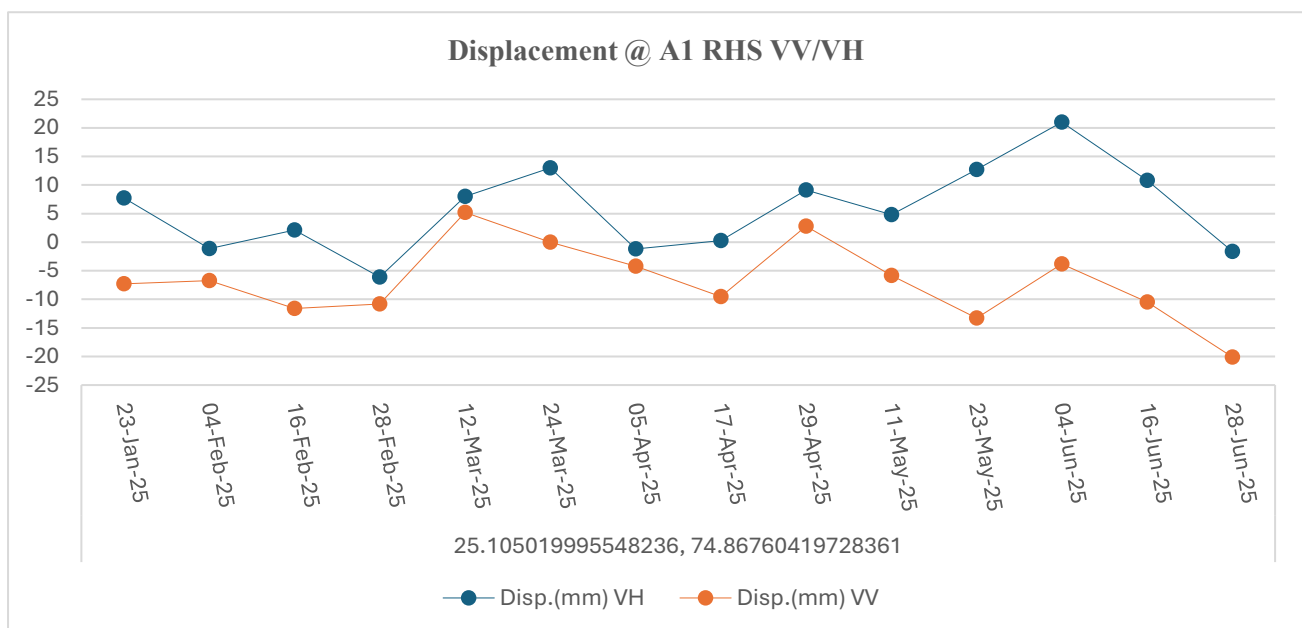
Critical Events & Anomalies:

- January 2025:** VH registers a high initial peak (+17 mm) while VV remains low (+3 mm). This marks the first instance of significant lateral movement without corresponding vertical displacement.
- June 2025:** This phase demonstrates opposing-direction movements, signalling strong environmental influence near the end of the monitoring cycle.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.105019995548236, 74.86760419728361)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows **significant positive variation**, fluctuating between **0 mm and +22 mm**, with multiple upward peaks
- **Vertical displacement (VV)** remains consistently negative, generally ranging from **-2 mm to -17 mm**, indicating continuous downward movement.

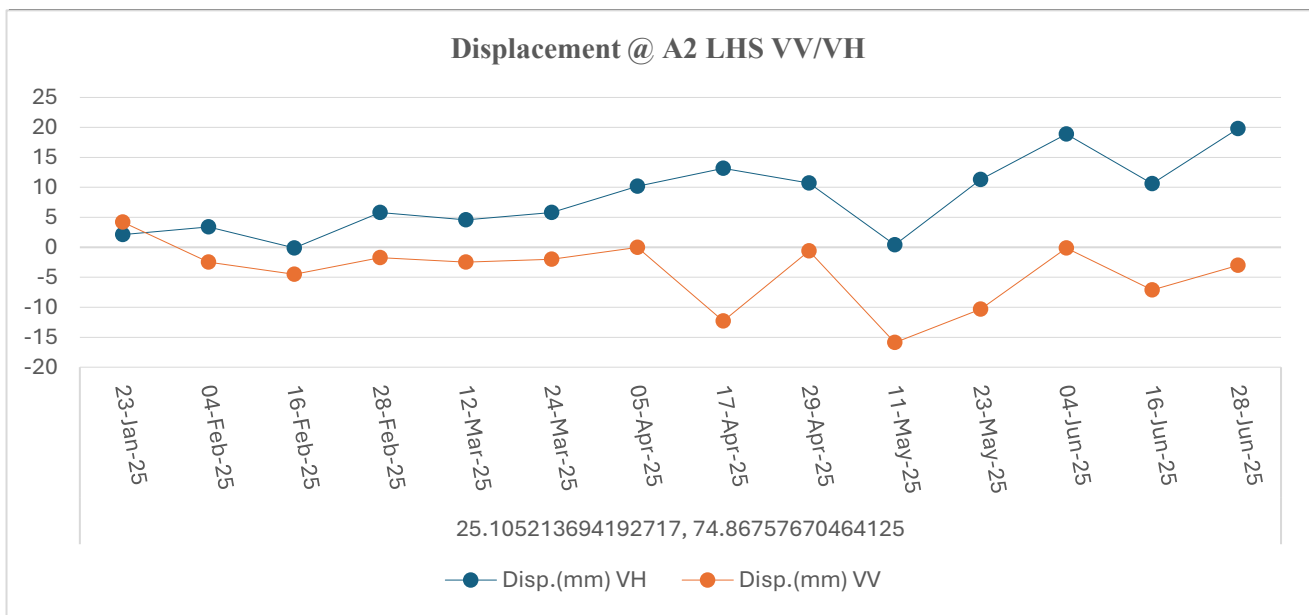
Critical Events & Anomalies:

- **March 2025:** VH increases sharply, peaking near +15 mm to +17 mm, while VV briefly rises toward 0 mm.
- **May 2025:** VV reaches one of its more negative values (-12 mm), while VH climbs to +11 mm, showing opposing-direction behaviour downward vertical movement alongside upward lateral displacement.
- **June 2025:** VH rises to its maximum peak (+22 mm on 04-Jun), indicating the strongest lateral movement in the dataset.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.105213694192717, 74.86757670464125)



Key Observations:

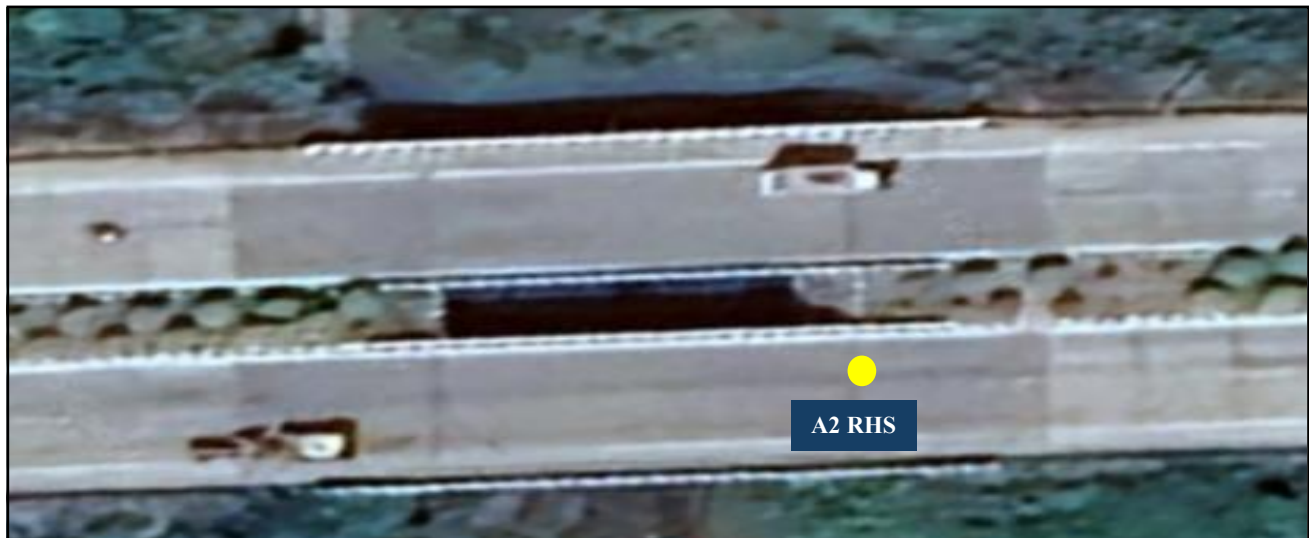
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows a clear upward trend, increasing from around +3 mm in late January to +22 mm by late June.
- **Vertical Displacement (VV)** remains consistently negative, ranging from approximately -2 mm to -15 mm, indicating ongoing downward settlement.

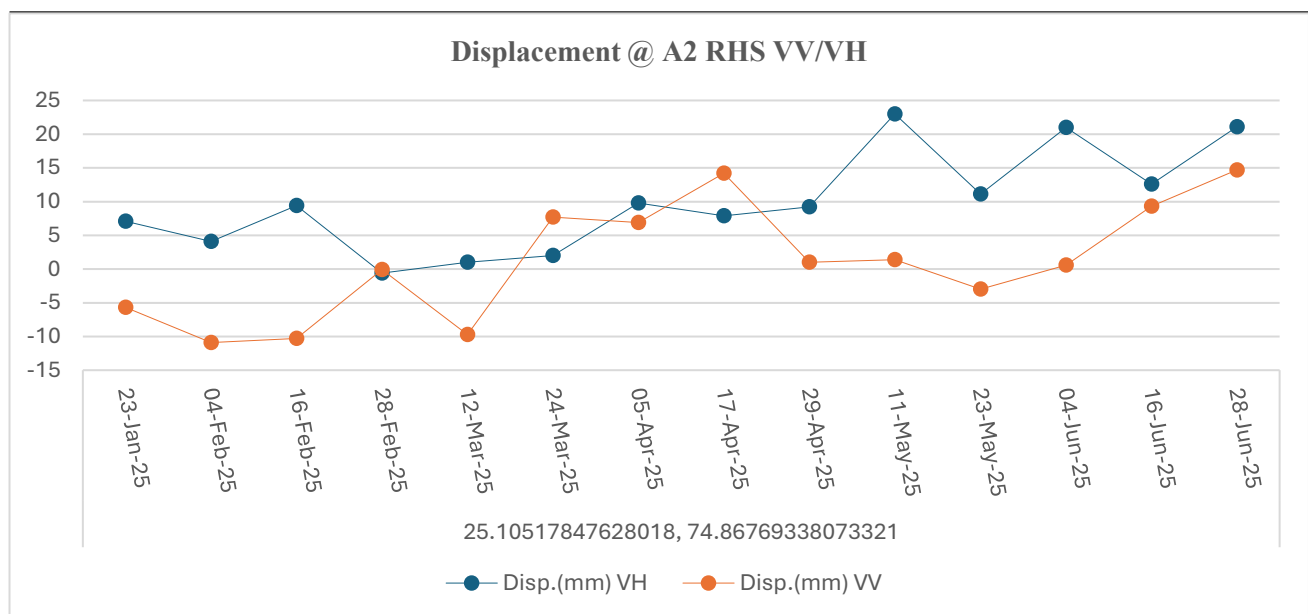
Critical Events & Anomalies:

- **April 2025:** VV drops sharply to around -14 to -15 mm, marking the largest negative vertical movement in the dataset.
- **June 2025:** VH reaches its overall maximum value (+22 mm), marking the strongest lateral displacement recorded.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.10517847628018, 74.86769338073321)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** starts negative (indicating initial downward movement) but shows a clear upward recovery trend, rising from around -10 mm in late January to approximately +15 mm by late June.
- **Horizontal (VH) displacement** exhibits a strong upward progression, increasing from roughly +5 mm in January to peaks of +20-22 mm by May and June 2025.

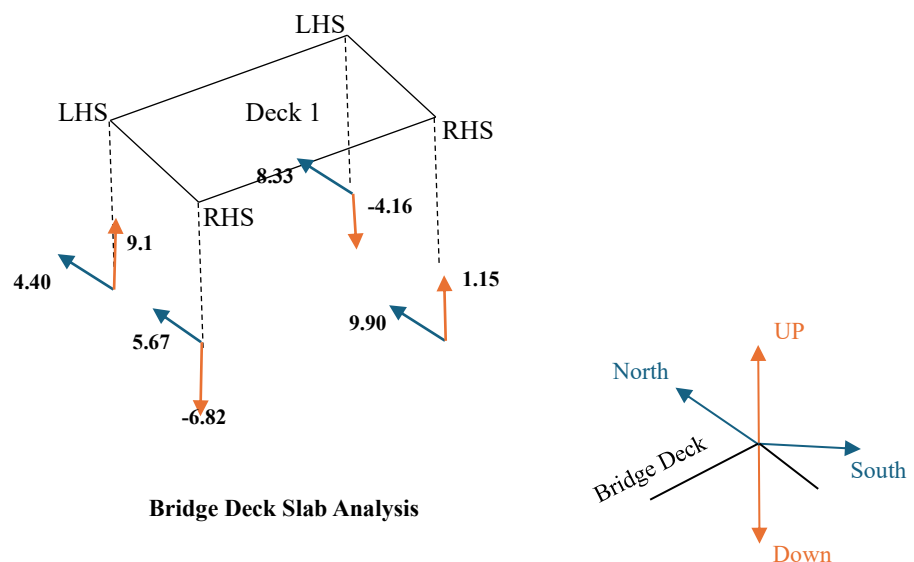
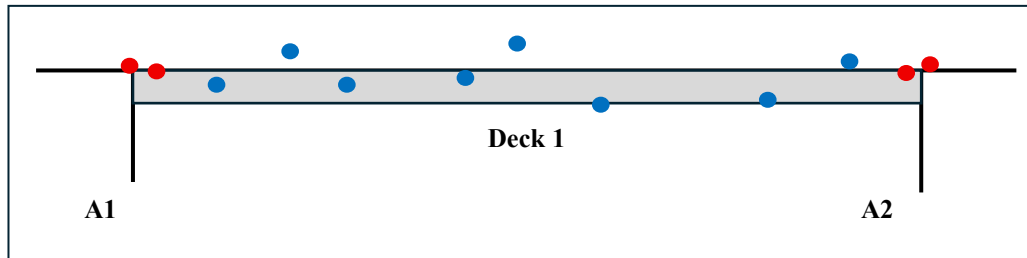
Critical Events & Anomalies:

- **April 2025:** VH records a strong upward shift to 11 mm, while VV reaches one of its early positive peaks (+6 mm).
- **May 2025:** VH spikes significantly to 22 mm, the highest horizontal movement up to that date.
- **June 2025:** represents the strongest combined deformation phase, with both vertical and horizontal movements trending upward significantly.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The magnitude of these movements remains within a normal operational range, suggesting that the slab is undergoing expected expansion-contraction behaviour without signs of overstress or distortion. The higher uplift readings near the left-hand side and moderate downward trend toward the centre indicate normal differential deck breathing, commonly observed in continuous slabs exposed to environmental gradients.

Horizontal Displacement Behaviour:

- Horizontal movements across the deck are generally modest, with values such as 4.40 mm, 5.67 mm, and 9.90 mm indicating minor to moderate northward shifts depending on the location.

CONCLUSION:

- The combined vertical and horizontal displacement patterns confirm that Deck 1 is performing well within acceptable structural behaviour limits, with no indicators of abnormal deformation or early-stage structural distress.
- No corrective actions are required at this stage, Routine monitoring should continue to ensure long-term stability and to detect any emerging deviations from these normal patterns.